CLAIMS

1. A Coexistence Device (1, 2, 3, 4, 5, 6, 6’), Cex (1, 2, 3, 4), for increasing the capabilities of an optical device, comprising:

- two or more first interfaces (11, 31, 51, 52) of the Cex for connecting to Optical Line Terminal, OLT, signal lines,
- a second interface (21, 61) of the Cex for connecting to a first Optical Distribution Network, ODN, signal line, and
- one or more third interface (41, 42, 71) of the Cex for connecting to one or more corresponding second Optical Distribution Network, ODN, signal line,
- a first optical filter (10, 302) configured for:
    - o passing through a first portion of a first filtered signal (110) of a first frequency band from a first OLT signal (108) on a first of the two or more first interfaces (11), and
    - o reflecting a second portion of the first filtered signal (310) and a first signal (101, 301) from the first OLT signal (108) on the first of the two or more first interfaces (11) of the Cex on to the second interface (21, 61) of the Cex,

and

- one or more third optical filters (30, 30’, 70, 90) configured for:
    - o receiving and passing from one or more second OLT signals (113, 123) from one or more corresponding second of the two or more first interfaces (31, 32) of the Cex of a second frequency band on to the one or more third interfaces (41, 42, 71) of the Cex.

2. The Cex (1, 2, 3, 4, 6, 6’) according to claim 1, wherein:
the first portion of the first filtered signal (110) is 100% of the first filtered signal (110), and the second portion of the first filtered signal (110) is 0% of the first filtered signal (110).

3. The Cex (1, 2, 3, 4, 6, 6’) according to claim 2, further comprising:

- a splitter (100,200) configured for splitting the first filtered signal (110) on a first interface (201) of the splitter (100,200) into:
    - o a first split signal (111, 114) on a second interface (202) of the splitter (100,200), and
    - o a second split signal (112, 115) on a third interface (203) of the splitter (100,200),
- a second optical filter (20, 60) configured for:
    - o receiving and passing from the first signal (101) being reflected by the first optical filter (10) from the first OLT signal (108), a signal of the second frequency band, on to the second interface (21, 61) of the Cex, and

- o receiving and reflecting the first split signal (111, 114) to the second interface (21, 61) of the Cex,

and

- the third optical filter (30, 70) is further configured for:
  - o receiving and reflecting the second split signal (112, 115) on to the third interface (41, 71) of the Cex.

4. The Cex (1, 2, 3, 4, 6, 6’) according to claim 3, wherein

the second (20, 60) and third (30, 70) optical filters are configured to reflect signals (111, 112, 114, 115) of the first frequency band from the second (21, 61) and third (41, 71) interfaces of the Cex respectively on to the second interface (202) and the third interface (203) of the splitter (100,200) respectively, and

the splitter (100, 200) is configured to pass the reflected signals (111, 112, 114, 115) on to the first interface (201) of the splitter (100,200) and on to the first of the two or more first interfaces (11) of the Cex via the first optical filter (10).

5. The Cex (1, 2, 3, 4, 6, 6’) according to claim 4, wherein

the second (20, 60) optical filter is further configured to pass signals of the second frequency band (101) from the second interface (21, 61) of the Cex, and further reflect these signals by the first optical filter (10) to the first of the two or more first interfaces (11) of the Cex.

6. The Cex (1, 2, 3, 4, 6, 6’) according to claim 5, wherein

the third (30, 70) optical filter is further configured to pass signals of the second frequency band (113)from the third interface (41, 71) of the Cex, on to the second of the two or more first interfaces (31) of the Cex.

7. The Cex (6, 6’) according to any one of claim 3 to 6, wherein

the Cex is arranged in two separate units (6, 6’) wherein:

- the first Cex unit (6) comprise:
  - o the first of the two or more first interfaces (11),
  - o the splitter (100,200),
  - o the second interface (21, 61) of the Cex, and
  - o the first and second optical filter (10, 20),

and

- the second portion of the Cex comprising:
  - o the second of the two or more first interfaces (31),

- o the third interface (41, 71) of the Cex, and
- o the third (90) optical filter,

the first Cex unit (6) further comprising a first intermediate interface (81) on the OLT side, and the second split signal (112, 115) is connected to the first intermediate interface (81), and

the second Cex unit (6’) further comprising a second intermediate interface (82) on the OLT side being connected to the consumer side of the third optical filter (90).

8. The Cex (1, 2, 3, 4) according to claim 7, further comprising:

a connector (83) for connecting the first intermediate interface (81) and the second intermediate interface (82).

9. The Cex (1, 2, 3, 4) according to any of the previous claims 3 to 8, further comprising:

- a fourth optical filter (50) configured for:
  - o passing a third OLT signal (109) of a third frequency band on a third of the two or more first interfaces (51) of the Cex and connecting to the splitter (100,200) side of the first optical filter (10) reflecting it to the first interface (201) of the splitter (100,200),

and

- the splitter (100,200) is configured for splitting the third OLT signal (109) into the first split signal (111’) and the second split signal (112’) on the second interface (202) and the third interface (203) of the splitter (100,200) respectively.

10. The Cex (1, 2, 3, 4) according to claim 9, wherein

the second (20) and third (30) optical filters are configured to reflect signals (111’, 112’) of the third frequency band from the second (21) and third (41) interfaces of the Cex on to the second interface (202) and the third interface (203) of the splitter (100,200) respectively, and

the splitter (200) is configured to pass the reflected signals (111’, 112’) on the first interface (201) of the splitter (100,200) and reflecting the signals (111’, 112’) of the third frequency band from the first optical filter (10, 302) and the fourth optical filter (50) passing this signal onto the third of the two or more first interfaces (51) of the Cex.

11. The Cex (1, 2, 3, 4) according to claim 10,

- the fourth optical filter is further configured for:

o reflecting a fourth OLT signal (119) of a fourth frequency band on a fourth of the two or more first interfaces (52) of the Cex to the splitter (100,200) side of the first optical filter (10) reflecting it to the first interface (201) of the splitter (100,200),

and

- the splitter (100,200) is configured for splitting the fourth OLT signal (119) on to the second (202) and third interface (203) of the splitter (100,200) respectively.

12. The Cex (1, 2, 3, 4) according to claim 11, wherein:

the second (20) and third (30) optical filters are configured to reflect signals (111’, 112’) of the fourth frequency band from the second (21) and third (41) interfaces of the Cex on to the second interface (202) and the third interface (203) of the splitter (100,200) respectively, and

the splitter (100, 200) is configured to pass the reflected signals (111’, 112’) of the fourth frequency band on the first interface (201) of the splitter (100,200) and reflecting the signals (111’, 112’) of the fourth frequency band from the first optical filter (10, 302) via the fourth optical filter (50) and onto the fourth of the two or more first interfaces (52) of the Cex.

13. The Cex (1, 2, 3, 4) according to any of the previous claims 3 to 8, further comprising:

- a third of the two or more first interfaces (51) of the Cex configured for: passing through, to a fourth interface (204) of the splitter (100,200), a third OLT signal (109) of a third frequency band on the third of the two or more first interfaces (51), and

- the splitter (100,200) is configured for splitting the third OLT signal (109) into the first split signal (114) and the second split signal (115) on the second interface (202) and the third interface (203) of the splitter (100,200) respectively.

14. The Cex (1, 2, 3, 4) according to claim 13, wherein

the second (60) and third (70) optical filters are configured to reflect signals (114, 115) of the third frequency band from the second (61) and third (71) interfaces of the Cex on to the second interface (202) and the third interface (203) of the splitter (100,200) respectively, and

the splitter (200) is configured to pass the reflected signals (114, 115) on the fourth interface (204) of the splitter (100,200) and onto the third (109) of the two or more first interfaces (51) of the Cex.

15. The Cex (1, 2, 3, 4) according to claim 14, further comprising:

- a fourth optical filter (50) configured for:

- o passing the third OLT signal (109) of the third frequency band on to the fourth interface (204) of the splitter (100,200),
- o reflecting a fourth OLT signal (119) of a fourth frequency band on a fourth of the two or more first interfaces (52) of the Cex to the fourth interface (204) of the splitter (100,200), and
- o the splitter (100,200) is configured for splitting the third frequency band (119) on to the second (202) and third interface (203) of the splitter (100,200) respectively.

16. The Cex (1, 2, 3, 4) according to claim 15, wherein:

the splitter (200) is configured to pass the reflected signals (114, 115) of the third frequency band on to the fourth interface (204) of the splitter (100,200) and onto the the fourth of the two or more first interfaces (52) of the Cex via the fourth optical filter (50).

17. The Cex (1, 2, 3, 4) according to claim 12 or 16, further comprising:

- further one or more optical filters (50m, 50n) configured for:
  - o passing through one or more further filtered signals of further frequency bands from further OLT signals (m, ...) on one or more further of the two or more first interfaces (51a) of the Cex, and
  - o reflecting even further signals from even further OLT signals (n, ...) on even further interfaces of the one or more first interfaces (52a) of the Cex, and
  - o the splitter (100,200) is configured for passing the further frequency bands on the second interface (202) and the third interface (203) of the splitter (100,200) respectively.

18. The Cex (1, 2, 3, 4) according to claim 17, wherein

the second (60) and third (70) optical filters are configured to reflect signals (111’, 112’, 114, 115) of one or more of the further frequency bands from the second (61) and third (71) interfaces of the Cex respectively on to the second interface (202) and the third interface (203) of the splitter (100,200) respectively, and

the splitter (200) is configured to pass the reflected signals (11’, 112’, 114, 115) on to the first interface (201) and fourth interface (204) respectively of the splitter (100,200) and to the first of the further two or more first interfaces (51a) of the Cex via the further one or more optical filters (50m, 50n).

19. The Cex (1, 2, 3, 4) according to claim 17 or 18, wherein:

the one or more optical filters (50m, 50n) are configured to reflect:

- the further frequency band of the passed signal to the further interfaces of the two or more first interfaces (51a) of the Cex, and/or
- the even further frequency bands of the further of the passed signal to the even further interfaces of the two or more first interfaces (52a) of the Cex.

20. The Cex (1, 2, 3, 4) according to any of the previous claims, wherein the first frequency band (110) is an XGS band.

21. The Cex (1, 2, 3, 4) according to any of the previous claims, wherein the first signal (101) is a GPON signal.

22. The Cex (1, 2, 3, 4) according to any of the previous claims, wherein the second signal (113) is a GPON signal.

23. The Cex (1, 2, 3, 4) according to any of the previous claims, wherein the first of the one or more first interface (11) is an XGS/GPON Combo port.

24. The Cex (1, 2, 3, 4) according to any of the previous claims, wherein the third of the one or more first interfaces (51) is a 25G/50G OLT Port.

25. The Cex (1, 2, 3, 4) according to any of the previous claims, wherein the fourth of the one or more first interfaces (52) are a UPG port.

26. A method for upgrading an Optical Line Terminal, OLT, combining two or more signal standards, the method comprising the steps:

- providing an OLT comprising two or more ports supporting a first of two or more Passive Optical Network, PON, standards, and at least two PON signal lines being connected to the PON side of the two or more ports, the two or more ports of the OLT further connects to a first and second Optical Distribution Network, ODN, signal lines on the consumer side of the two or more ports, and
- replacing one or more of any of the first of every second of the ports with a Cex (1, 2, 3, 4, 6, 6’) according to any one of claim 1 to 25.

27. The method according to claim 22, further comprising the step:

- o replacing the first of the at least two PON lines with a PON line providing a combination of a first of two or more standards and a second of the two or more standards,
- o connecting the PON line with the combination of the first of two or more standards to the first of the two or more first interfaces (11) of the Cex,
- o disconnecting the second of the at least two PON lines and connecting it to the second of the two or more first interfaces (31) of the Cex,
- o reconnecting the first and second ODN lines to the second interface (21, 61) of the Cex and the third interface (41, 71) of the Cex respectively,

and when the Cex is a Cex (6, 6') according to claim 7:

- o connecting the first intermediate interface (51) of the first Cex unit (6) with the second intermediate interface (61) of the second Cex unit (6').

28. The method according to claim 27, when the Cex is a Cex (6, 6') according to any one of claim 9 to 15, further comprising the step:

- connecting a PON line of a third of the two or more standards on the third of the two or more first interfaces (51) of the Cex.

29. The method according to claim 27 or 28, wherein:

the Cex (1, 2, 3, 4) is a Cex (1,2,3,4) according to any one of claim 11 to 19, and the method further comprising the step:

- connecting a PON lines of a fourth of the two or more standards on the fourth of the two or more first interfaces (52) of the Cex.

30. The method according to claim 29, further comprising the step:

- connecting a PON lines of a further of the two or more standards on the further of the two or more first interfaces (51a) of the Cex, and/or
- connecting a PON lines of an even further of the two or more standards on the even further of the two or more first interfaces (52b) of the Cex.